Vipransh Ojha

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Education

Vellore Institute of Technology — Bhopal, India
B.Tech in Computer Science and Engineering (AI & ML) — Sep 2023 – May 2027 (Expected)

Technical Skills

Languages: Python, C++, Java, Go, TypeScript, JavaScript, SQL, HTML5, CSS3
Backend & Cloud: FastAPI, Flask, Node.js, Express.js, REST APIs, PostgreSQL, MongoDB, AWS, Docker, Linux
AI/ML & Data: PyTorch, TensorFlow, scikit-learn, OpenCV, NLP, Pandas, NumPy, SciPy
Web3 & Tools: React, Solidity, Ethers.js, IPFS, Unity, PyOpenGL, Git

Experience

Software Development Intern — New Delhi, India
Government e-Marketplace (GeM), Ministry of Commerce & Industry — *Jun 2026 – Present*

- Built a reusable, multi-phase fraud-detection engine for GeM public procurement, combining MinHash candidate generation, geospatial/IP signals, lexical similarity, and graph analytics to flag suspected supplier collusion.
- Developed a repeatable price-audit workflow using IQR outlier detection, bill-of-materials cost adjustments, market/spec-equivalent fallbacks, and evidence-backed risk verdicts; passed 30 automated tests.
- Established recurring specification-matching workflows to compare external product feeds with GeM catalogues and generate auditable JSON/Excel reports for periodic procurement reviews.

Software Development Intern — New Delhi, India
BISAG-N (MeitY, Government of India) — *May 2025 – Jun 2025*

- Delivered a location-aware recommendation system for 500+ users, achieving 85% accuracy with content-based and SVD collaborative filtering.
- Engineered Flask/PostgreSQL backend services designed to handle 100+ queries per second under load.
- Implemented a JWT-secured interactive mapping interface, associated with a 25% increase in user engagement.

Research Intern – SERB-Sponsored Project — Bhopal, India
VIT Bhopal University — *May 2024 – Jun 2024*

- Developed *MolSpectra*, a molecular simulation and 3D visualization tool for a SERB-sponsored research project.
- Integrated 5+ quantum-chemistry packages, including GAMESS and Open Babel, to automate simulation workflows.
- Created a PyQt5/GLEW GUI for interactive molecular visualization and access to simulation tools.

Projects

ChronoVault – Decentralized Web3 Storage Protocol — Feb 2026 – Mar 2026

- Built a three-stage file pipeline with React, Go, AES-GCM encryption, IPFS/Pinata chunk storage, and Ethereum Sepolia hash anchoring.
- Added Solidity-based hash/timestamp records and Merkle-root verification to detect tampering during file retrieval.
- Integrated MetaMask/Ethers.js wallet access and file reconstruction using separate manifest and secret-key artifacts.

Dikastirio – AI-Assisted VR Courtroom — Nov 2025 – Jan 2026

- Prototyped a Unity VR courtroom for judges, counsel, and witnesses, with spatial presentation of legal evidence.
- Designed the AI agent *Themis* to coordinate 3 courtroom actions: calling witnesses, administering oaths, and recording adjournments.
- Mapped an agent workflow using Inya.ai/Gemma-3-27B, Gnani.ai transcription, and voice-driven display of PDFs, videos, and images.

Automated Dental Forensic Identification System — Aug 2025 – Oct 2025

- Trained a PyTorch Siamese network with pretrained EfficientNet-B0, generating 128-dimensional embeddings for dental-radiograph similarity scoring.
- Implemented balanced image-pair generation, 256×256 preprocessing, cosine-similarity scoring, and CUDA mixed-precision training.
- Prototyped on DENTEX, a dataset with 1,005 annotated panoramic X-rays, for automated dental-radiograph matching research.

Certifications & Achievements

- **NPTEL Elite Certifications:** Cloud Computing; Introduction to IoT (Top 2% Topper) – IIT Kharagpur.
- Ranked **Top 10** among 40,000+ registrations in the NASSCOM and Gnani.ai National Buildathon.
- **Founding Member & Student Coordinator, Agentic Intelligence Club**; solved 200+ LeetCode problems.